

UNIVERSITY OF EDINBURGH
SCHOOL OF INFORMATICS

INFR11287 ADVANCED TOPICS IN NATURAL LANGUAGE
PROCESSING

Mock Examination Paper 3

Multimodal Models, Agents, Interpretability, Parsing, and Ethics

Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

1. This is a practice paper, not an official examination paper.
2. Answer all questions. Different questions may have different numbers of total marks.
3. The paper is worth 50 marks in total.
4. Notes, calculators, and dictionaries are not permitted in the real examination.
5. Please answer concisely. Unless otherwise stated, do not write more than one or two sentences per mark assigned.

1. A museum wants to deploy a vision-language assistant. Visitors can take a photo of an exhibit or a label and ask questions such as “What object is this?”, “What does the label say?”, or “Why is this artefact important?” The museum wants the system to work for visually impaired visitors as well as general visitors.
 - (a) Explain why this task illustrates perceptually grounded NLP rather than a purely text-only language task. Your answer should refer to visual grounding or the symbol grounding problem. [3 marks]
 - (b) Describe the main components of a modern VLM built using a backbone LLM. Explain the role of the vision encoder and the projector or connector layer. [4 marks]
 - (c) The team can choose between a simple linear connector, an MLP connector, and an attention-based connector. Compare these choices in terms of training stability, representational power, and computational cost. [3 marks]
 - (d) Describe the kinds of training data the museum might need beyond generic image-caption pairs. Include at least one task-specific data type and one risk in collecting or using this data. [3 marks]
 - (e) Propose an evaluation plan for the deployed assistant. Your answer should distinguish intrinsic VLM evaluation from extrinsic task evaluation, and mention one accessibility or ethics concern. [2 marks]

2. The museum also wants an AI agent that can help staff book guided tours through several web forms. The agent may either use structured APIs or control a browser through screenshots, mouse clicks, and keyboard input. The museum is concerned about reliability, privacy, and irreversible mistakes.

- (a) Using the Russell and Norvig style definition, explain what makes this system an agent. Identify its sensors or observations and its actuators or actions in the two possible designs. [3 marks]
- (b) Compare a text-only tool-using agent with a vision-based computer-use agent for this booking task. Give one advantage and one disadvantage of each. [4 marks]
- (c) A vision-based computer-use agent is trained on trajectories of the form

$$I, s_0, a_0, s_1, a_1, \dots, s_T, a_T.$$

Explain what I , s_t , and a_t represent, and why trajectory history matters. [3 marks]

- (d) Describe two reliability or safety failure modes for this agent and one mitigation for each. At least one failure mode should involve the action space or UI state. [3 marks]
- (e) The museum asks whether the agent should be evaluated only by final task success rate. Explain why this is insufficient and name two additional quantities or behaviours that should be evaluated. [2 marks]

3. The following are multiple choice questions. For each question, choose the answer that is the most correct. Each question is worth 2 marks. [20 marks]

- (a) The connector in a visual-prefix VLM is necessary mainly because:
 - A. Image files are too large to store on disk.
 - B. Vision encoder features must be mapped into a representation compatible with the LLM.
 - C. The LLM must be removed before training.
 - D. It replaces tokenisation for all text inputs.
- (b) Increasing image resolution or decreasing patch size in a Vision Transformer usually:
 - A. Reduces the number of visual tokens.
 - B. Has no effect on computational cost.
 - C. Increases the number of visual tokens and therefore can increase attention cost.
 - D. Prevents the model from doing OCR.
- (c) In a computer-use agent, a termination action is important because:
 - A. The agent must know when to stop after achieving the goal.
 - B. It removes the need for observing the screen.
 - C. It makes all actions reversible.
 - D. It prevents the user from specifying a task.
- (d) A major difficulty for LLM watermarking in low-entropy decoding states is that:
 - A. There may be too few plausible token choices to impose a detectable watermark without hurting quality.
 - B. Low entropy always makes watermark detection perfect.
 - C. Watermarking only works for images.
 - D. The detector no longer needs a key.
- (e) Sparse autoencoders are useful in mechanistic interpretability because they can:
 - A. Map dense activations into a larger sparse set of candidate features.
 - B. Prove that every neuron is monosemantic.
 - C. Replace all evaluation benchmarks.
 - D. Eliminate the need for model activations.
- (f) Superposition refers to the idea that:
 - A. A model may represent more features than dimensions by overlapping features in activation space.
 - B. A parser must always output two trees.
 - C. All features correspond to single neurons.
 - D. Every attention head has the same function.
- (g) In a dependency parse, the main structure consists of:
 - A. Phrases nested inside larger phrases only.
 - B. Head-dependent relations between words.
 - C. A sequence of BPE merge operations.
 - D. A list of retrieved documents.
- (h) In constituency parsing by generating a linearised tree, a common validity problem is:
 - A. The model may produce unmatched or incorrectly typed brackets.
 - B. The model cannot output words.
 - C. Dynamic programming must be used at inference time.
 - D. The output cannot contain nonterminals.

- (i) Allocational harm differs from representational harm because allocational harm concerns:
 - A. Unequal distribution of opportunities or resources.
 - B. Only whether a stereotype appears in an embedding space.
 - C. Whether a model uses RoPE.
 - D. The number of tokens in a prompt.
- (j) A finite-state machine is limited, compared with richer formalisms, because it:
 - A. Cannot represent all regular languages.
 - B. Has no finite set of states.
 - C. Does not have unbounded stack-like memory for nested dependencies.
 - D. Is identical to a Transformer with full attention.